SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Code of Conduct:

I Dhivya Dharshini S *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S Dhivya

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Problem Identification	20%	1	1	1	1	1
Debugging	25%	2	1	1	1	2
Optimization	20%	1	1	1	1	1
Analysis	20%	1	1	1	1	1
Code Quality	15%	1	1	1	1	1

24/30

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks /**26**, /**27**, and /**28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	**Next Hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

Answer 1

1. Problem identification

- **Given:** IP 192.168.10.65, mask 255.255.255.192 (/26), gateway 192.168.10.1. The workstation should talk to hosts in 192.168.10.0/26 but cannot.
- **Suspected root cause:** the IP address and the gateway are not in the same subnet, and the IP is not in the intended subnet.

2. Debugging

Block size = 256 − 192 = 64. The last octet of the IP is ANDed with the mask: 65 = 01000001, 192 = 11000000, so 01000001 AND 11000000 = 01000000 = 64.

Item	Subnet of the workstation IP	Subnet of the gateway and target hosts
Address checked	192.168.10.65	192.168.10.1
Subnet	192.168.10.64/26	192.168.10.0/26
Network address	192.168.10.64	192.168.10.0
Broadcast address	192.168.10.127	192.168.10.63
Valid host range	192.168.10.65 – 192.168.10.126	192.168.10.1 – 192.168.10.62

Validity check:

- 192.168.10.65 is a valid host address (the first host of 192.168.10.64/26), but it belongs to the wrong subnet.
- 192.168.10.1 is a valid host address, but it belongs to 192.168.10.0/26, not to the workstation's subnet.

Configuration errors:

- 1. The workstation IP is in 192.168.10.64/26, so it is not on the same subnet as the hosts in 192.168.10.0/26. Traffic to them needs a router.
- 2. The default gateway 192.168.10.1 is outside the workstation's own subnet, so the workstation cannot reach the gateway directly. Without a reachable gateway it cannot use the router.

3. Optimization (corrected configuration)

Setting	Wrong value	Corrected value
IP address	192.168.10.65	192.168.10.10 (any free address from .2 to .62)
Subnet mask	255.255.255.192	255.255.255.192 (unchanged)
Default gateway	192.168.10.1	192.168.10.1 (unchanged, now in the same subnet)

Cisco Packet Tracer commands for the router interface that serves this LAN:

```
Router(config)# interface gigabitEthernet0/0
Router(config-if)# ip address 192.168.10.1 255.255.255.192   ! gateway of
192.168.10.0/26
Router(config-if)# no shutdown
```

On the PC (Desktop > IP Configuration, static): IP 192.168.10.10, mask 255.255.255.192, gateway 192.168.10.1.

Usable hosts after correction: $2^6 - 2 = 64 - 2 =$ **62** (61 for workstations once the gateway takes one address).

4. Analysis

- Check: 10 lies between 0 and 63, so 192.168.10.10 is in 192.168.10.0/26. It is neither the network address (.0) nor the broadcast address (.63).
- The gateway .1 is now in the same subnet as the workstation, so ARP works and the workstation can reach both local hosts and the router.
- Test with ping 192.168.10.1 and then a ping to another host in the subnet.
- Alternative: if the workstation really belongs in 192.168.10.64/26, keep .65 and move the gateway into that subnet (for example 192.168.10.126 on the router interface).

1. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

Answer 2

1. Problem identification

- **Given:** Device A 2001:db8::ff00:42:8329/64 and Device B 2001:db8:0:0:1::1/64. They are on the same network, but communication fails.
- The task is to find whether a prefix or addressing mismatch causes this.

2. Debugging

Expansion of both addresses:

Group	1	2	3	4	5	6	7	8
Device A	2001	0db8	0000	0000	0000	ff00	0042	8329
Device B	2001	0db8	0000	0000	0001	0000	0000	0001

- A: :: stands for 8 − 5 = 3 zero groups. Full form: **2001:0db8:0000:0000:0000:ff00:0042:8329**.
- B: :: stands for 8 − 6 = 2 zero groups. Full form: **2001:0db8:0000:0000:0001:0000:0000:0001**.

Network prefix (/64 = first 4 groups):

- Device A: 2001:0db8:0000:0000, so **2001:db8::/64**.
- Device B: 2001:0db8:0000:0000, so **2001:db8::/64**.

Same subnet? Yes. Both addresses have the same first 64 bits and the same /64 length.

Addressing or prefix mismatch: there is **none** in the prefix. The value 1 in the fifth group of Device B looks like part of the network, but with /64 it falls in the interface ID (bits 65 to 80). It would only change the subnet if it were in the fourth group (2001:db8:0:1::1 is a different /64). So the addresses are valid and in the same subnet, and the fault must be elsewhere.

Other causes to check:

- Interface down, or the devices are on different VLANs or switch ports.
- IPv6 not enabled or not set to static on a Packet Tracer PC.
- ICMPv6 or Neighbor Discovery blocked by a firewall or ACL.
- A mistyped prefix length on one device, which would give different subnets.

3. Optimization (clean configuration)

Device	Optimized address	Prefix
Device A	2001:db8::a/64	2001:db8::/64
Device B	2001:db8::b/64	2001:db8::/64

Router interface on this LAN, if used (Packet Tracer):

```
Router(config)# ipv6 unicast-routing
Router(config)# interface gigabitEthernet0/0
Router(config-if)# ipv6 address 2001:db8::1/64   ! gateway for the /64
Router(config-if)# no shutdown
```

Host addresses in the corrected /64: host bits = 128 − 64 = 64, so 2^{64} = **18,446,744,073,709,551,616** addresses.

4. Analysis

- Short, readable interface IDs (::a, ::b) avoid the mistake of placing a digit in the wrong group.
- Verify with ipv6 config or show ipv6 interface brief, then ping 2001:db8::b from Device A. In Wireshark, filter icmpv6 to see Neighbor Solicitation and Echo packets.
- IPv6 has no broadcast, so the full 2^{64} range is available to hosts, apart from the reserved subnet-router anycast address (all-zeros interface ID).

2. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

Answer 3

1. Problem identification

- **Given:** 192.168.1.0/24 split into departments with the masks /26, /27 and /28. Some departments run out of addresses while others have many unused ones.
- The question does not give host counts, so this answer **assumes** these departmental needs: Admin 20 hosts, Sales 45 hosts, IT 12 hosts. The method works for any other numbers.
- **Assumed current plan:** Admin /26, Sales /27, IT /28, placed in that order.

2. Debugging

Dept	Subnet	Network	Host range	Broadcast	Usable	Needed	Result
Admin	192.168.1.0/26	192.168.1.0	.1 – .62	192.168.1.63	62	20	42 unused (68% wasted)
Sales	192.168.1.64/27	192.168.1.64	.65 – .94	192.168.1.95	30	45	**15 short**
IT	192.168.1.96/28	192.168.1.96	.97 – .110	192.168.1.111	14	12	2 unused

- Usable addresses = 2^h − 2. Allocated: 62 + 30 + 14 = 106 usable. Needed: 20 + 45 + 12 = 77.
- Unused block outside the subnets: 192.168.1.112 – 192.168.1.255 (144 addresses).

Allocation mistakes:

- 1. The largest mask (/26) was given to the department that needs the fewest hosts (Admin).
- 2. Sales, the biggest department, got a /27 and cannot fit 45 hosts.
- 3. The masks were assigned by order, not by the number of hosts needed.

3. Optimization (VLSM)

Rule: pick the smallest subnet where 2^h − 2 ≥ hosts needed, and allocate the biggest first.

Dept	Hosts needed	Subnet chosen	Network	Host range	Broadcast	Usable	Spare
Sales	45	/26 (62)	192.168.1.0	.1 – .62	192.168.1.63	62	17
Admin	20	/27 (30)	192.168.1.64	.65 – .94	192.168.1.95	30	10
IT	12	/28 (14)	192.168.1.96	.97 – .110	192.168.1.111	14	2

Cisco Packet Tracer interface commands (one router interface per department):

```
Router(config)# interface gigabitEthernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.192    ! Sales gateway
Router(config)# interface gigabitEthernet0/1
Router(config-if)# ip address 192.168.1.65 255.255.255.224   ! Admin gateway
Router(config)# interface gigabitEthernet0/2
Router(config-if)# ip address 192.168.1.97 255.255.255.240   ! IT gateway
```

4. Analysis

- Utilization improves from 32% (Admin), 150% (Sales, overloaded) and 86% (IT) to 67%, 73% and 86%. Every department now fits.
- The blocks stay on their boundaries (0, 64, 96) and do not overlap, and 192.168.1.112 – 192.168.1.255 is still free for growth or /30 router links.
- If a department grows, resize its subnet by the same rule instead of adding a new one.

3. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

Answer 4

This question repeats Question 2 with the same addresses, so the answer is the same. It is restated here in short so that this question can be marked on its own.

1. Problem identification

- Device A is 2001:db8::ff00:42:8329/64 and Device B is 2001:db8:0:0:1::1/64. Both are said to be on the same network, but they cannot communicate.

2. Debugging

Item	Device A	Device B
Expanded address	2001:0db8:0000:0000:0000:ff00:0042:8329	2001:0db8:0000:0000:0001:0000:0000:0001
Network prefix (/64)	2001:db8::/64	2001:db8::/64
Interface ID	0000:ff00:0042:8329	0001:0000:0000:0001

- The prefixes are identical, so both devices are in the **same subnet**. There is no prefix mismatch.
- The digit 1 in the fifth group of Device B belongs to the interface ID at /64. It is not part of the network prefix.
- The failure therefore comes from outside the prefix: link or VLAN problems, IPv6 not enabled on an interface, ICMPv6 or Neighbor Discovery blocked, or a wrongly typed prefix length.

3. Optimization

- Assign clear addresses in the same /64: Device A 2001:db8::a/64 and Device B 2001:db8::b/64, with the router interface at 2001:db8::1/64.
- Host addresses in the corrected subnet: 2^(128 − 64) = 2^{64} = **18,446,744,073,709,551,616**.

4. Analysis

- Check with show ipv6 interface brief and ping 2001:db8::b. Use the Wireshark filter icmpv6 to confirm that Neighbor Solicitation and Echo messages are exchanged.
- Using short interface IDs prevents a digit from landing in the wrong group.

4. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.

- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

Answer 5

1. Problem identification

- The router table is: 192.168.1.0/24 via 10.0.0.2, 192.168.2.0/24 via 10.0.0.3, default route via 10.0.0.1.
- Packets for 192.168.2.100 are not reaching the intended network. The task is to check the lookup, find the next hop and locate the fault.

2. Debugging

Longest-prefix match for 192.168.2.100:

Route	Prefix length	Does 192.168.2.100 match?
192.168.1.0/24	24	No: third octet 2 ≠ 1
192.168.2.0/24	24	**Yes**: 192.168.2.100 AND 255.255.255.0 = 192.168.2.0
Default (0.0.0.0/0)	0	Yes, but only a /0, the shortest match

- The matching entry is **192.168.2.0/24**, because /24 is longer than /0.
- Next hop selected: **10.0.0.3**.
- Subnet check: 192.168.2.0/24 has hosts 192.168.2.1 – 192.168.2.254 and broadcast 192.168.2.255, so 192.168.2.100 is a valid host in the intended subnet.

Configuration errors: the table itself is consistent, so the lookup is correct. The problem lies in the forwarding path to 10.0.0.3:

- 1. The next hop 10.0.0.3 may be unreachable (interface down, or no connected route to the 10.0.0.0 network on this router).
- 2. The next hops 10.0.0.2 and 10.0.0.3 may be swapped, so 192.168.2.0/24 is sent to the wrong router.
- 3. The router at 10.0.0.3 may lack a route to 192.168.2.0/24 or a route back to the source.

3. Optimization

Correct and document the static routes, and check the next hops (Packet Tracer):

```
Router(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.2    ! customer
network 1
Router(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.3    ! customer
network 2
Router(config)# ip route 0.0.0.0 0.0.0.0 10.0.0.1              ! default route
Router# show ip route
Router# ping 10.0.0.3
Router# traceroute 192.168.2.100
```

- Keep the two /24 routes as they are, because their next hops differ and a summary would send traffic to the wrong router.
- If both networks later use the same next hop, replace them with one summary route (192.168.0.0/22 would also cover 192.168.0.0/24 and 192.168.3.0/24, so use it only if that is intended).
- If a route points to a next hop that can fail, add a backup route with a higher administrative distance.

4. Analysis

- The traffic for 192.168.2.100 uses 192.168.2.0/24 via 10.0.0.3 and not the default route, so the router's decision is right and the fix is on the next-hop side.
- **Route used if no match exists:** the default route 0.0.0.0/0 via **10.0.0.1**.
- Verify with show ip route, a ping to the next hop and Wireshark on the 10.0.0.x link to see whether the packets leave the router and whether replies come back.

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